

International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in CSE Midterm Examination, Autumn 2025
Course Code: STAT 2311 Course Title: Probability and Statistics
Total marks: 30, Time: 1 hour 30 minutes

[Answer all the questions; Figures in the right-hand margin indicate full marks.]

CO DL

1(a) Define Statistics as a subject and explain its connection with Computer Science and Engineering. 4 CO1 C3

1(b) Define a variable. Mention its types. Classify the scale of measurement of each of the following terms: Download speed, Network protocol used, Performance rating, IQ scores, distance traveled, and Proficiency in coding. 6 CO1 C4

2(a) Explain the objectives of using measures of central tendency in data preprocessing. Discuss with an example when the geometric mean cannot be calculated. 4 CO1 C3

2(b) A database administrator is monitoring the response times of queries in a database system. The response times (in milliseconds) are shown into the following frequency table: 6 CO1 C6

Response Time (ms)	5–10	10–15	15–20	20–25	25–30
Frequency	3	7	13	5	2

(i) Calculate the median, third quartile, and 70th percentile from the given data.

(ii) Draw an appropriate graph to locate 1st quartile and 85th percentile.

3(a) What are the various types of measures of dispersion in statistics? “Relative measures of dispersion are more suitable for comparing two or more data sets” – why? 4 CO1 C2

3(b) The exam scores of five students in class A and class B are given below: 6 CO1 C3

Student	1	2	3	4	5
Class A	50	55	60	65	65
Class B	20	40	50	70	80

(i) Which class has the higher average exam score?

(ii) Which class of students has more consistent exam scores?

Or

a) Explain the following terms with example (any two):
 (i) Population, (ii) Primary data, (iii) Questionnaire 4 CO1 C2

b) The following data represents the number of children in seven families: 2, 0, 5, 3, 4, 6, 1. Calculate the arithmetic mean and standard deviation for the above dataset. What happens to your measures if a fixed number, such as 2, is added to the number of children in each family? 6 CO1 C3

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Midterm Examination, Spring- 2025
Course Code: STAT 2311 Course Title: Probability and Statistics
Total marks: 30, Time: 1 hours 30 minutes

[Answer all the questions; Figures in the right hand margin indicate full marks.]

- | | | CO | DL |
|--|---|-----|----|
| 1. | | | |
| a) What is a questionnaire? | 2 | CO1 | C1 |
| b) Design a structured questionnaire with at least 20 questions on the topic: "Factors Affecting Adverse Reactions of COVID-19 Vaccine in Bangladesh." | 8 | CO1 | C6 |
| 2. | | | |
| a) What is the geometric mean? How is it different from the harmonic mean? Explain the applications of various measures of central tendency. | 5 | CO1 | C2 |
| b) The boot time (in seconds) of 20 different systems in a lab is recorded as follows: | 5 | CO1 | C4 |

Boot Time (sec)	5-10	10-15	15-20	20-25	25-30
No. of Systems	2	5	7	4	2

(i) Calculate the first quartile (Q_1), ninth decile (D_9), and eighty-seventh percentile (P_{87}) from the given data.

(iv) Draw an appropriate cumulative frequency curve (ogive) and use it to locate the values of Q_2 , D_9 , and P_{71} .

- | | | | |
|---|---|-----|----|
| 3. | | | |
| a) What are the various types of measures of dispersion in statistics? Explain any two with appropriate examples. | 5 | CO1 | C2 |
| b) The accuracy rates (in percentage) of two machine learning models, Model A and Model B are as follows: | 5 | CO1 | C4 |

Model A	89	92	85	91	88	90
Model B	86	88	87	84	89	Your weight (in kilograms)

(i) Which model has a higher average accuracy?

(ii) Which model demonstrates greater consistency in its accuracy rates?

Or

- | | | | |
|--|---|-----|----|
| a) Define a variable with an example, and explain how a discrete variable differs from a continuous variable. | 5 | CO1 | C2 |
| b) Write your birthdate in ddmmyyyy format (with a zero before single-digit day or month if needed), separating each digit with commas. Then, calculate the coefficient of variation for these digits. | 5 | CO1 | C4 |

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Spring- 2024

Course Code: STAT 2311 Course Title: Probability and Statistics

Total marks: 30, Time: 1 hours 30 minutes

[Answer all the questions; Figures in the right hand margin indicate full marks.]

CO DL

- 1.
- a) Define statistics and explain its importance in the field of computer science and engineering. 5 CO1 C2
- b) What is a quantitative variable? Determine whether these variables are discrete or continuous or qualitative? 5 CO1 C5
- (i) Time it takes for students to finish their assignment (ii) Number of books in the library (iii) Price of laptop (iv) Number of visitors to hit a website (v) Thesis titles.

- 2.
- a) Define weighted average. Can you compare the weighted average with other types of averages? Write some real-world applications where weighted average is commonly used? 5 CO1 C2
- b) In a study, students were surveyed regarding their weekly coding practice time, which was categorized as follows: 5 CO1 C4

Hours Spent (per week)	0-5	5-10	10-15	15-20	20-25	25-30
Frequency	5	8	9	20	12	6

- (i) Calculate the mean and median weekly coding practice time (ii) What percentage of students spends more than 20 hours per week on coding practice? (iii) Draw an ogive curve and hence locate median.

Or

- a) Distinguish between: (i) Primary data and secondary data.
(ii) Discrete variable and continuous variable.
- b) Suppose the heights in inches of the students in your section are as follows: 66, 77, 68, 63, 72, 60, 66, 70, 71 and your height in inches. What would be the mean, median and mode of this data? Can you compute the variance, standard deviation, and coefficient of variation for this dataset? Which measure would best represent this data?
- 3.
- a) What is meant by dispersion? What are its important measures? Discuss mean deviation and the coefficient of variation. 5 CO1 C2
- b) The runs scored by two batsmen in their last 6 ODI cricket matches are given below: 5 CO1 C4

Batsman- X	65	25	78	Sum of last two digits of your ID	51	34
Batsman-Y	40	35	18	28	25	41

- (i) Which batsman has a higher average run?
(ii) Which batsman is more consistent?

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Autumn- 2023

Course Code: STAT 2311 Course Title: Probability and Statistics

Total marks: 30, Time: 1 hours 30 minutes

[Answer all the questions; Figures in the right hand margin indicate full marks.]

CO DL

- 1.
- a) Define statistics as a subject and explore its connection with data science. 4 CO1 C2
- b) What is the difference between discrete and continuous variables? Identify whether these variables are discrete/continuous or qualitative? 6 CO1 C5
- (i) The temperature of a CPU or a server (ii) The amount of memory being used by a process (iii) Program execution time (iv) The count of clicks on a IIUC webpage (v) The programming language being used in a project.

- 2.
- a) Explain the procedure for calculating the geometric mean, harmonic mean, and median of ungrouped data using illustrative examples. Write down the formulae for computing Q_3 , D_8 and P_{87} from a grouped frequency distribution. 4 CO1 C2
- b) A database administrator is monitoring the response times of queries in a database system. The response times are grouped into intervals (in milliseconds) as shown in the frequency table below: 6 CO1 C4

Response Time (ms)	0-10	10-20	20-30	30-40	40-50
Frequency	3	8	12	5	2

Calculate (i) the mean and median response time for the queries (ii) 1st quartiles, 6th deciles and 78th percentiles (iii) Draw an ogive curve and hence locate median and third quartile.

- 3.
- a) Define mean deviation and coefficient of variation (CV). Discuss how measures of dispersion can help identify code sections that exhibit inconsistent execution times, aiding developers in optimizing software performance. 4 CO1 C2
- b) A software company is assessing the following performance of two programmers, Shafeen and Shahariar in terms of the number of lines of code they write per hour. 6 CO1 C4
- Shafeen's coding speed over five days: 130, 135, 125, 140, 150
- Shahariar's coding speed over five days: 120, 125, 130, 135, 150
- Calculate the coefficient of variation for both Shafeen and Shahariar and determine who has a more consistent coding speed.

Or

- a) Explain the following statistical terms with example (Any two): 4 CO1 C2
- (i) Population (ii) Variable (iii) Secondary data (iv) Questionnaire
- b) In a software development team, the time (in hours) taken to complete code reviews was recorded: 8, 5, 9, and 7 6 CO1 C5
- Using the data justify: (i) $A.M \leq G.M \leq H.M$ (ii) $A.M \times H.M = GM^2$

International Islamic University Chittagong
 Department of Computer Science and Engineering
 B. Sc. in CSE Midterm Examination, Spring- 2023
 Course Code: STAT 2311 Course Title: Probability and Statistics
 Total marks: 30, Time: 1 hours 30 minutes
 [Answer all the questions; Figures in the right hand margin indicate full marks.]

CO DL

1. a) Explain the concept of variable with its importance in statistical research. 5 CO1 C4
 Classify the following variables as either qualitative or quantitative: Ethnicity, Number of siblings, Country of origin, Volume, Brain activity as measured via EEG.

- b) Distinguish between: (i) Primary data and secondary data. 5 CO1 C4
 (ii) Discrete variable and continuous variable.

2. a) Under what circumstances is it not possible to calculate the geometric mean? Describe the concept of weighted arithmetic mean with its application. 5 CO1 C2

- b) The following grouped frequency table shows the length of time, t , in minutes, visitors watched an octopus swimming around a tank at an aquarium. 5 CO1 C5

Time(t)	$0 < t \leq 5$	$5 < t \leq 10$	$10 < t \leq 15$	$15 < t \leq 20$	$20 < t \leq 25$	$25 < t \leq 30$
Visitors	4	3	7	5	2	1

Determine graphically the median and mode of the time visitors spent observing the octopus.

3. a) What do you mean by dispersion? Indicate the different measures of dispersion. Distinguish between mean deviation and standard deviation. 5 CO1 C4

- b) Following is the frequency distribution of life length in hours of 20 electric bulbs: 5 CO1 C4

Life Length	8.5-13.5	13.5-18.5	18.5-23.5	23.5-28.5	28.5-33.5
Frequency	3	5	7	4	1

Calculate the coefficient of variation (CV) and the standard error of mean (SEM) using the above data and comment on your results.

Or

- a) Suppose your midterm test score is 73 and your final exam score is 85. Using weights of 30% for the midterm and 70% for the final exam, compute the weighted average of your scores. If the minimum average for an A is 75, will you earn an A? 5 CO1 C4

- b) Using the numbers 2, 4, 8, 5, and 7, provide a justification for the following relationships: (i) $A.M \geq G.M \geq H.M$ (ii) $\sigma_x > MD_x$ 5 CO1 C5

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Autumn- 2022

Course Code: STAT 2311 Course Title: Probability and Statistics

Total marks: 30, Time: 1 hours 30 minutes

[Answer all the questions; Figures in the right hand margin indicate full marks.]

CO DL

1.

- a) Define Statistics. How is probability and statistics used in computer engineering? 4 CO1 C2
- b) Suppose a researcher wants to find the average systolic blood pressure of the employees of a big firm. For this purpose, a sample of 80 employees has been selected randomly from that firm and their systolic blood pressure measurements have been obtained. 6 CO1 C6
- (i) Identify the population, sample, variable, and data for the above example.
- (ii) What kind of variable is "systolic blood pressure"?
- (iii) Distinguish between discrete and continuous variable.

2.

- a) What do you mean by measures of central tendency? Describe the method of determining geometric mean and harmonic mean. 4 CO1 C4
- b) The following data shows the yearly temperature of north Bengal of Bangladesh: 6 CO1 C3

Temperature (°C)	5-10	10-15	15-20	20-25	25-30	30-35
# of days	2	4	11	7	4	1

- (i) Find average yearly temperature (ii) Compute the modal temperature.
- (iii) Using the cumulative frequency curve, determine the median temperature.

3.

- a) Explain mean deviation and standard deviation. Show that the standard deviation of first n natural number is $\sqrt{\frac{n^2-1}{12}}$ 4 CO1 C4

deviation of first n natural number is $\sqrt{\frac{n^2-1}{12}}$

- b) The lives (in hours) of 50 randomly selected flashlight batteries are: 6 CO1 C5

Class	6.95 to 7.45	7.45 to 7.95	7.95 to 8.45	8.45 to 8.95	8.95 to 9.45	9.45 to 9.95
Interval	7.45	7.95	8.45	8.95	9.45	9.95
Frequency	2	10	13	11	5	4

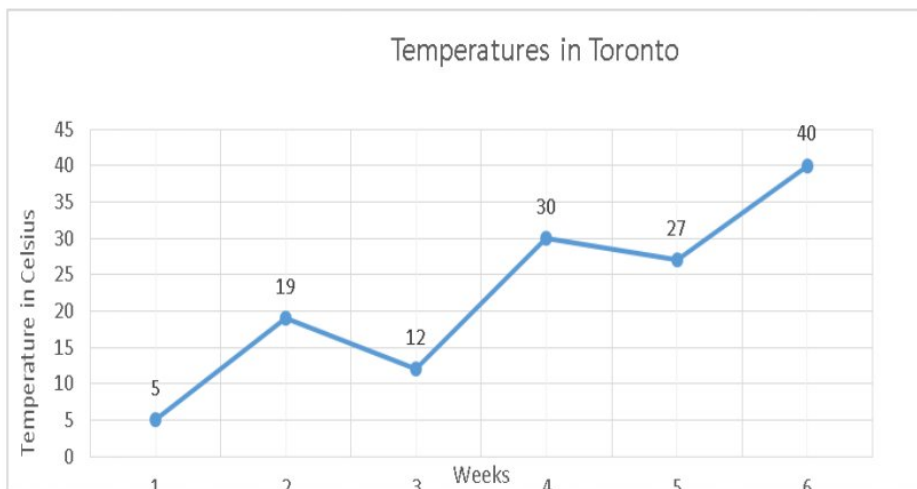
Using the above data verify that $\sigma > 1.0$.

Or

- a) Write your birthday into ddmmYYYY format and separated each digit by commas. Compute the coefficient of variation of these digits. 4 CO1 C4
- b) Find the arithmetic mean and standard deviation of the first n natural number whose frequencies are equal to the corresponding numbers. 6 CO1 C5

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Midterm Examination, Spring-2022
Course Code: STAT 2311 Course Title: Probability and Statistics
Total marks: 21, Section: 3AM
Time: 2 hours 30 minutes for exam + 30 minutes for submission
 [Answer all the questions; Figures in the right hand margin indicate full marks.]

- | | | CO | DL |
|---|----------|------------|-----------|
| 1. | | | |
| a) Statistics play an intrinsic role in computer science and vice versa-Explain. | 3 | CO1 | C6 |
| b) Why is a sample used more often than a population? Determine whether the data set is a population or a sample. Explain your reasoning. | 2 | CO1 | C5 |
| (i) The BMI of each person.
(ii) The speed of every fifth car passing a police speed trap.
(iii) The annual salary for each employee at a company.
(iv) A study of 1501 infants in Bangladesh was conducted to find a link between heart rhythm abnormality and sudden infant death syndrome.
(v) A survey of 1104 households in a certain region found that 65% subscribe to cable television. | | | |
| c) Answer the questions as it pertain to the following graph. | 2 | CO1 | C6 |



- | | | | |
|--|----------|------------|-----------|
| (i) What is the range of the values on the horizontal scale?
(ii) How many points are on the graph?
(iii) What is the highest temperature recorded?
(iv) At what point did the temperature dip? | | | |
| 2. | | | |
| a) Explain median graphically. Discuss its difference from the mean. In which situations is the median more effective to describe the center of the data? | 3 | CO1 | C3 |

- b) If the arithmetic mean of the data given below is 28, Calculate (i) The missing frequency, and (ii) The median and (iii) the mode of the series. 4 CO1 C6

Profit (in TK.):	0-10	10-20	20-30	40-50	50-60	60-70
No. of Firms:	12	18	27	X	17	6

Also draw an ogive curve and hence locate median.

- 3.
- a) What are the differences between central tendency and dispersion? 2 CO1 C3
- b) The grouped frequency table shows the length of service in years of employees who have been working for a company for at least ten years. Calculate an estimate of the mean deviation and standard deviation of the length of service of these employees. Verify that $\sigma_x > MD_{\bar{x}}$. 3 CO1 C5

Length of Service (x)	$10 \leq x < 15$	$15 \leq x < 20$	$20 \leq x < 25$	$25 \leq x < 30$	$30 \leq x < 40$
Frequency (f)	5	Last two digit of your ID	12	3	2

- c) Suppose the heights in inches of the students in your class are as follows: 66, 68, 63, 60, 62, 64, 64, 65, 66, 66, 66, 66, 68, 68, 69, 70, 71, 72, 72, 71, 73, and **your height in inches**. What would be the mean of this data? How about the median and mode? Would you be able to calculate the variance for this data? How about the standard deviation? 2 CO1 C4

THE END